EXPERIMENT: 6

A PYTHON PROGRAM TO DO FACE RECOGNITION USING SVM CLASSIFIER.

Aim:

To implement face recognition using Support Vector Machine (SVM) classifier.

Procedure:

1. Import required libraries.
2. Load face dataset (Olivetti dataset).
3. Split dataset into training and testing sets.
4. Train the SVM classifier.
5. Predict test data using the trained model.
6. Evaluate accuracy.

CODE:

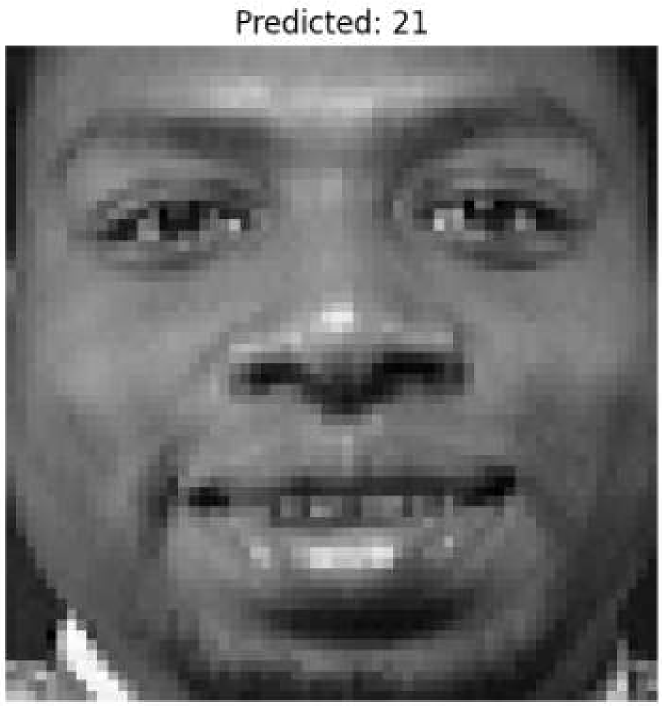
```
!pip install scikit-learn
```

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.datasets import fetch_olivetti_faces

from sklearn.svm import SVC

from sklearn.model_selection import train_test_split

from sklearn.metrics import accuracy_score
```

```
faces = fetch_olivetti_faces()

X = faces.data      # flattened images
y = faces.target    # labels (0–39)

X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.2, random_state=42
)

model = SVC(kernel='linear')
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))

for i in range(5):
   plt.imshow(X_test[i].reshape(64,64), cmap='gray')
   plt.title(f"Predicted: {y_pred[i]}")
   plt.axis('off')
   plt.show()
```

OUTPUT:

Accuracy: 0.975



RESULT:

The SVM classifier successfully recognized faces with good accuracy.